

Manuscripts considered — osteosarcoma-mets-dll3-h7r2

Master inventory: every paper reviewed in this case

Generated 2026-05-05

Manuscripts considered — osteosarcoma-mets-dll3-h7r2

Master inventory: 8 clinical (5 included, 3 considered & excluded) + 5 pre-clinical (4 included, 1 considered & excluded) + 2 additional trial publications/registrations. One row per manuscript, sorted by year (newest first). Excluded rows are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Report	Type	Inclusion	Indication / model	Design	n	Effect size	Variance	Toxicities (type, n/N, rate)	Case fit	Notes	
(2025) — N Engl J Med	PMID clinical	included	tarlatamab (DLL3 × CD3 BiTE)	SCLC after platinum-based chemo (2L); or SCLC; ECOG 0-1	randomized phase 3	509	0.6 HR — OS (HR vs chemo)	95% CI 0.47–0.77; 95% CI; median OS 13.6 vs 8.3 mo; p=<0.001	CRS (n=254, 56%); CRS G>=3 (n=254, 1%); ICANS-like neurologic event (n=254, 12%); treatment-related AE G>=3 (n=254, 24%); neutropenia G>=3 (n=254, 3%)		Confirmatory phase-3 OS benefit in SCLC. Establishes mechanism convincingly; cross-tumor extrapolation to osteosarcoma still requires DLL3 IHC confirmation.
—/— (2025) — record / ASCO 2025 abstract TPS2700	trial	included	tarlatamab (DLL3 × CD3 BiTE)	advanced solid tumors (basket (2L+); DLL3 IHC ≥25% (stage 1) or ≥1% (stage 2))	single-arm phase 2 basket (Simon two-stage)	29	— — ORR at 18 months (RECIST 1.1)	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
—/— (2025) — record	trial	included	+ MAP	newly diagnosed osteosarcoma (localized + metastatic) (1L); unselected	phase 2/3 RCT + MAP vs MAP)	0	— — EFS	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

(2023) — N Engl J Med	PMID 37861218	clinical	included	tarlatamab (DLL3 × CD3 BiTE)	previously treated SCLC (2L+); SCLC after platinum-based therapy and ≥1 prior line; ECOG 0-1; brain mets allowed if treated	open-label phase 2	220	40 % — ORR (BICR)	95% CI 29–52; 95% CI in 10mg cohort	CRS (n=100, 51%); CRS G≥3 (n=100, 1%); ICANS-like neurologic event (n=100, 10%); treatment-related AE G≥3 (n=100, 30%); treatment discontinuation due to AE (n=100, 3%)	Pivotal trial driving FDA accelerated approval. Cross-tumor row: patient has osteosarcoma, not SCLC — included as the load-bearing efficacy evidence the basket trial NCT06788938 extrapolates from.
Zhang/Shi (2023) — Aging (Albany NY)	PMID 37179118		included	DLL3 pan-cancer expression atlas	TCGA pan-cancer transcriptomic + protein-atlas analysis	of DLL3 expression across tumor types and correlations with TMB, MSI, and tumor	33	moderate — DLL3 expression detectable across 18 cancer types; co-varies with immune infiltration in subsets. Sarcoma signal present but not	vs GTEx normal-tissue baseline; translatability: low	n/a (preclinical)	weak Bulk RNA-level analysis. Surface-protein expression — the requirement for BiTE/ADC engagement — is not interrogated. Useful for hypothesis generation; insufficient to gate clinical decisions.

Yao/Pavel (2022) — Oncologist	PMID 35983951	included	DLL3 as therapeutic target — review	Literature review	overview of DLL3 biology, IHC and landscape (BiTEs, ADCs, CAR-T,	—	moderate — DLL3 is a antigen with low normal-tissue expression — attractive when present at adequate density. detection (e.g. SP347 IHC) is the consensus gating biomarker for DLL3-directed therapies.	vs —; translatability: med	n/a (preclinical)	weak	Focuses on neuroendocrine neoplasms. Does not establish osteosarcoma-specific evidence. Reinforces the protein-IHC requirement that the user's 'DLL3 RNA' alone does not satisfy.
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(2021) — JTO	PMID 33508456	clinical	excluded — Drug by sponsor (AbbVie withdrew Rova-T after the TAHOE phase-3 SCLC trial showed worse OS vs topotecan). Mechanism	—	—	—	—	—	—	—	Excluded: Drug discontinued by sponsor (AbbVie withdrew Rova-T after the TAHOE phase-3 SCLC trial showed worse OS vs topotecan). Mechanism (DLL3-targeted PBD-warhead ADC) is informative for the target rationale but the molecule itself is not procurable.
Italiano/Blay (2020) — Lancet Oncol	PMID 32078813	clinical	included	advanced osteosarcoma after standard therapy (2L+); Advanced osteosarcoma (n=42) and Ewing sarcoma cohorts; ECOG 0-1; adequate organ function	single-arm phase 2 (CABONE)	42	11.9 % — ORR + 6-mo	95% CI 4.0–25.6; 95% CI; 6-mo 33.3%	hypertension G>=3 (n=42, 19%); hand-foot reaction G>=3 (n=42, 14%); hypophosphatemia G>=3 (n=42, 12%); transaminitis (ALT/AST elevation) G>=3 (n=42, 12%); thromboembolism (3/42, 7%); any treatment-related AE G>=3 (n=42, 68%)	strong	Pre-specified osteosarcoma cohort. Supports current AOST2032 phase-2/3 frontline trial. Reasonable 2L+ option in non-trial setting.

Hipp/Wolf (2020) — Clin Cancer Res	PMID 32694156		excluded — Original BiTE construct (AMG 757 / tarlatamab) in SCLC models — superseded by Giffin 2021 which includes the cytolysis data directly relevant to the IHC ≥1% / ≥25% threshold this case turns on.	tarlatamab (DLL3 × CD3 BiTE)	—	—	—	—	—	n/a (preclinical)	—	Excluded: Original BiTE construct (AMG 757 / tarlatamab) characterization in SCLC models — superseded by Giffin 2021 (pmid:35983951) which includes the cytolysis data directly relevant to the IHC ≥1% / ≥25% threshold this case turns on.
Davis/Maki (2019) — JCO	PMID 31013172	clinical	included	regorafenib	metastatic osteosarcoma (2L+); Metastatic osteosarcoma after at least one prior line of chemotherapy; ECOG 0-1; adequate organ function	RCT (SARC024)	42	3.6 months — median PFS	95% CI 2.0–7.6; vs 1.7 mo placebo; HR 0.42 (95% CI 0.21-0.85); p=0.017	hand-foot reaction G≥3 (n=22, 23%); hypertension G≥3 (n=22, 14%); fatigue G≥3 (n=22, 14%); mucositis G≥3 (n=22, 9%); any treatment-related AE G≥3 (n=22, 70%); treatment-related death G5 (0/22, 0%)	strong	Small RCT with crossover; OS not improved (crossover diluted).

(2019) — Lancet Oncol	PMID 30477937	clinical	included	regorafenib	metastatic osteosarcoma (2L+); Adult metastatic osteosarcoma after conventional chemo failure; French Sarcoma Group cohort	RCT	38	65 % — at 8 weeks	95% CI 47–80; 95% CI	hand-foot reaction G>=3 (n=26, 23%); fatigue G>=3 (n=26, 15%); hypertension G>=3 (n=26, 12%); transaminitis (ALT/AST elevation) G>=3 (n=26, 8%); any treatment-related AE G>=3 (n=26, 76%); dose reduction for AE (n=26, 65%)	strong	Independent French replication of SARC024. Two-RCT convergence is the strongest 2L+ osteosarcoma evidence base.
(2019) — Clin Cancer Res program)	—		included	tarlatamab (DLL3 × CD3 BiTE)	DLL3-positive SCLC xenograft + humanized PBMC co-culture	Bispecific antibody co-engages DLL3 on tumor cells and CD3 on T cells, redirecting cytotoxic T-cell killing.	n=8 in n=3	strong — cytolysis demonstrated; ≥1% surface DLL3 by IHC sufficient for activity in SCLC models. Tumor regression sustained beyond treatment cessation in xenograft.	vs vehicle + PBMCs without bispecific; translatability: med	n/a (preclinical)	All published mechanistic data are SCLC-derived. No primary preclinical osteosarcoma data with tarlatamab in the public literature. Translatability to osteosarcoma depends on whether DLL3 is membrane-localized and density-sufficient in osteosarcoma cells — currently unknown.	

Wang/Zhu (2018) — J Exp Clin Cancer Res	PMID 29475441	included	pathway in (target rationale)	Osteosarcoma cell lines (MG-63, U2OS, Saos-2); primary osteosarcoma samples	axis implicated in stemness and SOX2OT lncRNA upregulates signaling.	n=3 per	moderate — EGCG + doxorubicin combination suppresses Notch3/DLL3 signaling and reduces stemness in suggests pathway is operative, not just expressed.	vs vehicle + scrambled siRNA / parental cell line; translatability: low	n/a (preclinical)	weak	Pathway-level mechanistic study; does NOT measure cell-surface DLL3 protein expression by IHC. Does not address whether DLL3 is on the membrane (required for BiTE/ADC engagement) vs intracellular only. The user's 'DLL3 RNA expression' input matches the type of evidence here — but this work cannot bridge the gap that DLL3-targeted therapies need.
(2017) — Clin Cancer Res	PMID 28289205	clinical	excluded — activity in requires / BRCA-like phenotype; patient profile.json shows no NGS evidence of HRD. Out of scope absent testing.	olaparib —	—	—	—	—	—	—	Excluded: PARP-inhibitor activity in osteosarcoma requires HR-deficiency / BRCA-like phenotype; patient profile.json shows no NGS evidence of HRD. Out of scope absent confirmatory testing.

(2017) — Lancet Oncol	PMID 28499700	clinical	excluded — SARC028	—	—	—	—	—	—	—	Excluded: SARC028 osteosarcoma cohort: ORR 5% (1/22), no durable responses; weak signal does not justify a recommendation in a 2L+ DLL3-targetable case where carry stronger published RCT evidence.
			cohort: ORR 5% (1/22), no durable responses; weak signal does not justify a in a 2L+ case where carry stronger published RCT evidence.								